FEYMAN

FMI IMU

Software Integration Guide

IM19 / IM23 / IM26 series inertial measurement units

Document Version: V1.5

Applicable Models: IM19, IM23, IM26 series

Interface: UART, AT command set

www.feymani.com

youtube.com/@FeymanIncFMI

github.com/Feyman-Technology

Revision History

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CHANGES IN V1.5 Layout redesigned to the FMI application-note style. New Figure 1 (lever arm) and Figure 2 with Table 2 (upright and inverted mounting, installation angle and lever-arm signs). Clarified that `LEVER_ARM2` is always entered along the module's default axes. Receiver types and working modes presented as tables; tilt modes 2 and 3 (408 / 410) merged as equivalent settings with a video demo link; AHRS working mode removed.

NOTE This document describes the software configuration and message protocols of FMI IMU modules operating with GNSS aiding, and applies to the IM19, IM23 and IM26 series. Commands and protocols depend on the firmware actually loaded in the module; confirm the firmware version with `AT+VERSION` before production.

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1 Introduction

FEYMAN (FMI) IMU modules — including the IM19, IM23 and IM26 series — are pure inertial sensors containing only an accelerometer and a gyroscope. An IMU can operate on its own (as an AHRS, for example), but for applications such as tilt survey or agricultural machine control, GNSS aiding is required to achieve the best performance. With GNSS aiding, tactical-grade IMU performance can be reached. This document describes the software configuration and message protocols of a GNSS-aided IMU.

Before operating a GNSS-aided IMU, make the following preparations.

1. Configure the GNSS board to output position and velocity information to the IMU, for example NMEA GGA, RMC and GST.
    a. For Hemisphere RTK boards, BIN3 is recommended; it is sufficient on its own and can replace NMEA.
    b. For the u-blox F9P, high-precision mode must be enabled in the NMEA output configuration.
2. Connect the PPS signal of the GNSS to the IMU and make sure the active PPS edge matches the IMU configuration.
3. Check the synchronisation status of the IMU via the STATUS word or the `AT+CHECK_SYNC` command.
4. Set the installation angle of the IMU and the coordinates of the antenna phase centre in the IMU coordinate frame.
5. Set the working mode of the IMU; different modes require different alignment initialisation procedures.
    a. FMI IMUs support several functions such as tilt survey, agricultural navigation and vehicle dead reckoning. Each function has its own working-mode parameter, which must be set correctly.
6. Configure the output messages of the IMU, such as the NAVI and Euler angle messages.
7. Perform the required calibration to achieve the best accuracy.

2 Configuration Commands

The IMU module is configured over a serial interface. Commands have the following structure:

```
AT+COMMAND[=][PARA1],[PARA2],[PARA3]\r\n
```

All letters of a command are upper case; parameters in [] are optional. Unless explicitly saved, the configuration is lost after a restart. The default baud rate is 115200 bps. The basic response to a command is:

```
\r\nOK\r\n\r\n        or        \r\nError\r\n\r\n
```

2.1 Command Overview

Command	Function
AT+SYSTEM_RESET	System reset
AT+SAVE_ALL	Save parameters
AT+UPDATE_APP	Enter firmware update mode
AT+GNSS_CARD=OEM	Set the GNSS RTK receiver type
AT+READ_PARA=SYSTEM/ALL	Read out parameters
AT+LOAD_DEFAULT	Load default parameters
AT+AUTO_FIX=ENABLE/DISABLE	Installation misalignment angle estimation
AT+CLUB_VECTOR=X,Y,Z	Set the vector that maps the position onto the pole tip
AT+NAVI_OUTPUT=UART1,ON/OFF/RATE	Set the output of the binary NAVI message
AT+MEMS_OUTPUT=UART1,ON/OFF/RATE	Set the output of the binary MEMS message
AT+GNSS_OUTPUT=UART1,ON/OFF	Set the output of the binary GNSS message
AT+NASC_OUTPUT=UART1,ON/OFF	Set the output of the ASCII NAVI message
AT+EULER_OUTPUT=UART1,ON/OFF/RATE	Set the output of the Euler angle message
AT+NAVA_OUTPUT=UART1,ON/OFF/RT	Set the output of the NAVA message for AR applications
AT+NAVV_OUTPUT=UART1,ON/OFF/RT	Set the output of the NAVV message for image surveying
AT+SET_CAMERA_TRI=5/10/20	Set the camera trigger rate of the VM23-IS
AT+LEVER_ARM2=X,Y,Z	Set the lever arm in the standard coordinate frame
AT+CHECK_SYNC	Query whether time synchronisation with the GNSS is complete
AT+HIGH_RATE=ENABLE/DISABLE	Set the high-rate mode
AT+ACTIVATE_KEY=KEY	Module activation
AT+ALIGN_VEL=1.0	Set the velocity threshold for initial alignment

Command	Function
AT+VERSION	Query the firmware version
AT+GNSS_PORT=PHYSICAL_UART2	Set the port for the GNSS receiver; effective after saving and restarting
AT+WORK_MODE=X	Set the working mode of the module
AT+INSTALL_ANGLE=X,Y,Z	Set the installation angle of the module
AT+THIS_PORT	Query the number of the serial port
AT+FILTER_STOP=ENABLE/DISABLE	Put the filter into stop mode or bring it back
AT+FILTER_RESET	Filter reset
AT+CHECK_CRC=N	Calculate the firmware CRC for verification, where N is the firmware size
AT+OUTPUT_DISABLE=UARTx	Switch off the output of all messages on serial port x
AT+CALIBRATE_MODE=STEP1/STEP2	Factory calibration command
AT+SET_PPS_EDGE=RISING/FALLING	Set the PPS edge; by default the falling PPS edge is used
AT+CAN_MSG=RATE,MSG	Set the CAN output messages and rate
AT+CAN_CFG=BPS,STD/EXT,ID	CAN configuration
AT+VEHICLE_CALI	Start the calibration for DR mode

Table 1. AT command set. Availability depends on the firmware loaded in the module; confirm with AT+VERSION.

2.2 Commands in Detail

(1) AT+GNSS_CARD=OEM

Sets the type of the GNSS RTK receiver. There are four types: Hemisphere boards, NOVATEL boards, UNICORE boards and generic OEM boards.

Setting	Receiver	Messages that must be fed in
AT+GNSS_CARD=HEMI	Hemisphere	Bin3
AT+GNSS_CARD=UNICORE	Unicore	BESTPOSB, PSRVELB and GPGGA
AT+GNSS_CARD=OEM	Generic OEM	GPGGA, GPRMC and GPGST
AT+GNSS_CARD=NOVATEL	NovAtel	BESTPOSB, BESTVELB and GPGGA; the velocity must be set to Doppler velocity

NOTE Position data at a rate of at least 5 Hz is required. The RTK receiver must supply a velocity derived from Doppler. For HEMI, UNICORE, NOVATEL and all GNSS types other than OEM, the leap-second time compensation in the NAVI position data protocol is 18 s.

(2) AT+AUTO_FIX=ENABLE

In tilt survey applications, this command triggers the calibration of the installation misalignment angle.

> **NOTE** It is only effective when the filter has converged and is stable, that is, in the Ready state. Once the estimation is complete, the setting returns automatically to DISABLE.

(3) AT+CLUB_VECTOR=X,Y,Z

This command maps the position of the GNSS antenna phase centre onto any specified point, using the current attitude. For this mapping, the origin of the coordinate frame lies at the antenna phase centre and the axis directions correspond to the IMU coordinate frame.

(4) AT+LEVER_ARM2=X,Y,Z

The GNSS antenna phase centre does not coincide with the origin of the MEMS sensor coordinate frame. In motion, the positions and velocities measured by the GNSS therefore differ from those of the MEMS sensor; a mapping is required to compensate for this.

As shown in Figure 1, the three-dimensional coordinates of the GNSS antenna phase centre in the IMU coordinate frame are (X, Y, Z). Use this command to set the lever arm of the IMU; the more accurate the lever arm, the higher the accuracy the IMU can achieve.

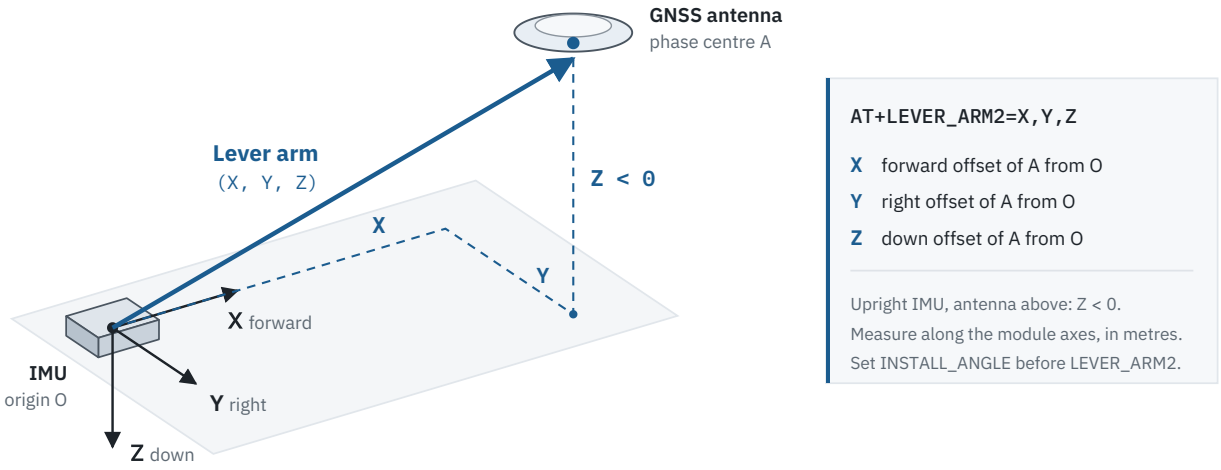


Figure 1. Lever arm: the position of the GNSS antenna phase centre A relative to the IMU origin O, resolved along the IMU axes X (forward), Y (right) and Z (down).

> **NOTE** A. For an upright module, if the antenna is above the IMU, Z is negative; otherwise Z is positive.
>
> B. X, Y and Z are always measured along the IMU's default (module) axes, whether the module is mounted upright or inverted; the installation angle must be set before the lever arm.

Inverted mounting. If the module is assembled upside down — on the underside of the PCB or on the back of the GNSS receiver — send `AT+INSTALL_ANGLE=180,0,0` first, but still measure the lever arm along the module's own default axes. Inverted, those axes point Y left and Z up, so the signs change: an antenna above the IMU now gives a positive Z, and an antenna to the right gives a negative Y. The module uses the installation angle to apply the lever arm correctly.

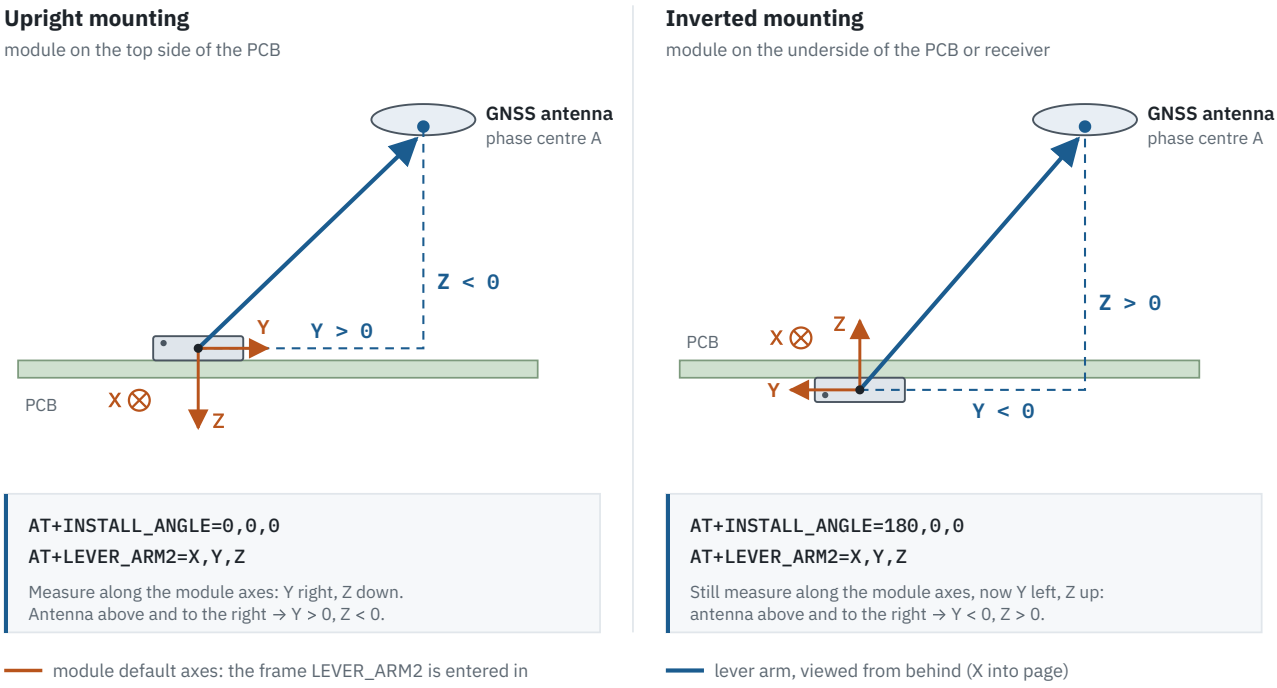


Figure 2. Upright and inverted mounting, viewed from behind with X pointing into the page. `LEVER_ARM2` is always entered along the module's default axes, so inverting the module reverses the signs of Y and Z.

The module can be turned over in two ways, and each needs its own installation angle. Table 2 lists both, with the lever-arm signs that result for an antenna mounted ahead of, to the right of and above the IMU.

Mounting	Module axes point	Installation angle	Lever arm signs
Upright	X forward, Y right, Z down	`AT+INSTALL_ANGLE=0,0,0`	`X>0 Y>0 Z<0`
Inverted, rolled over X turned over side to side	X forward, Y left, Z up	`AT+INSTALL_ANGLE=180,0,0`	`X>0 Y<0 Z>0`
Inverted, pitched over Y turned over front to back	X backward, Y right, Z up	`AT+INSTALL_ANGLE=0,180,0`	`X<0 Y>0 Z>0`

Table 2. Installation angle for upright and inverted mounting. Lever arm signs are for an antenna ahead of, to the right of and above the IMU, measured along the module's default axes.

(5) `AT+WHEEL_ARM=X,Y,Z`

In agricultural heading mode this command is used to compensate for the mounting position. Its parameters are the coordinates of the vehicle's centre of rotation in the IMU coordinate frame, in metres (m).

On a front-steered agricultural machine, for example, the centre of rotation lies directly above the rear wheels. If the device is mounted well away from this centre of rotation — more than about 0.5 m horizontally —

compensation with this command is recommended.

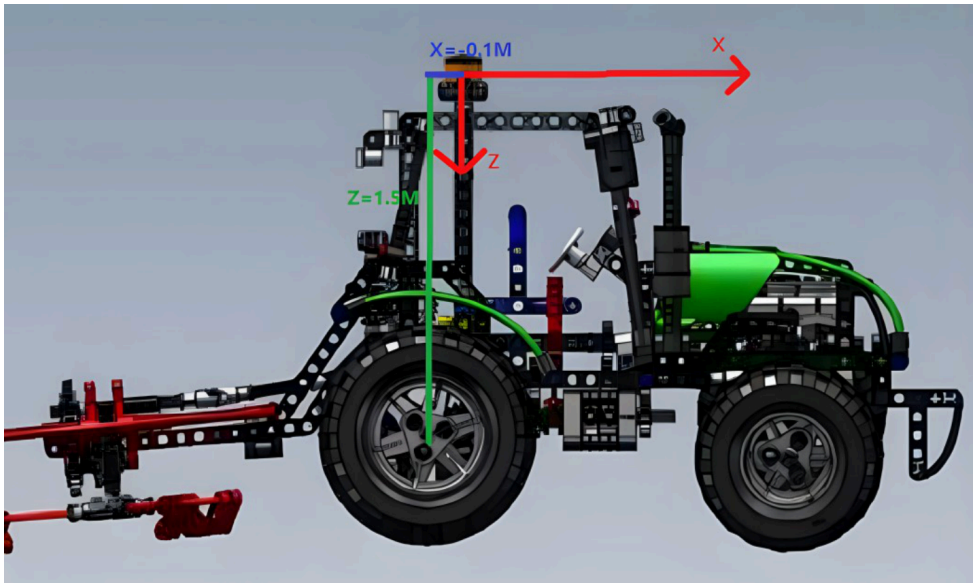


Figure 3. Wheel arm: the centre of rotation above the rear axle, expressed in the IMU frame (here X = −0.1 m, Z = 1.5 m).

Configuration example:

```
AT+WHEEL_ARM=-0.1,0,1.5
```

(6) AT+HIGH_RATE=ENABLE

High-rate mode outputs NAVI data at 100 Hz and suits highly dynamic applications with low latency. In this mode accuracy may be reduced in favour of dynamic performance, so enable it only when real-time behaviour is critical.

(7) AT+FILTER_STOP=ENABLE

This command puts the IMU module into a quiescent mode that prevents filter initialisation. If the IMU filter is already running, it is reset and then halted. The function is useful when the module has to wait for a particular user input before normal operation.

(8) AT+WORK_MODE=X

Configures the working mode of the module. The default is tilt mode 1.

Mode	X	Use and initialisation
Tilt mode 1	152	Default. Hold the pole tip fixed on the ground and swing the RTK pole back and forth to initialise the tilt survey.
Tilt mode 2 / 3	408 / 410	Equivalent settings. Initialise by swinging the RTK pole back and forth, or simply walking, or simply hand-holding the device and moving it. See the video demo: youtube.com/watch?v=cB1ThtNHu2Q
Vehicle mode	13	In-vehicle dead reckoning (DR) applications.
Heading mode	72	Agricultural machine automation.
Excavator mode	1033	Machine control applications on excavators.

Table 3. Working modes set with AT+WORK_MODE=X.

3 Message Protocols

All binary messages share the same frame: a three-byte header `fmi`, a one-byte data type, the payload, a 16-bit checksum (the byte sum of everything before it) and the tail `ed`. Offsets below are 1-based byte positions.

3.1 MEMS Raw Data

The MEMS raw data uses little-endian byte order. Acceleration values are given in g, gyro values in rad/s.

Field	Data	Type	Offset
Header	`f`	`char`	1
Header	`m`	`char`	2
Header	`i`	`char`	3
Data type	`m`	`char`	4
Field 1	`HHMMSS.SS`	`double`	5
Field 2	Acceleration X axis	`float`	13
Field 3	Acceleration Y axis	`float`	17
Field 4	Acceleration Z axis	`float`	21
Field 5	Gyro X axis	`float`	25
Field 6	Gyro Y axis	`float`	29
Field 7	Gyro Z axis	`float`	33
Field 8	Reserved	`float`	37
Field 9	Reserved	`float`	41
Field 10	Reserved	`float`	45
Checksum	Sum of bytes 1–48	`uint16_t`	49
Tail	`e`	`char`	51
Tail	`d`	`char`	52

Table 4. MEMS raw data message, data type `m`, 52 bytes.

3.2 GNSS Raw Data

The GNSS raw data uses little-endian byte order. It originates from the output of the GNSS receiver, and its rate matches the receiver's output rate. If GNSS raw data is required in real time, request it directly from the GNSS receiver.

Field	Data	Type	Offset
Header	`f`	`char`	1
Header	`m`	`char`	2
Header	`i`	`char`	3
Data type	`g`	`char`	4
Field 1	`HHMMSS.SS`	`double`	5
Field 2	Latitude (degrees)	`double`	13
Field 3	Longitude (degrees)	`double`	21
Field 4	Height	`double`	29
Field 5	North velocity	`float`	37
Field 6	East velocity	`float`	41
Field 7	Vertical velocity (down)	`float`	45
Field 8	Variance of north position	`float`	49
Field 9	Variance of east position	`float`	53
Field 10	Variance of height position	`float`	57
Field 11	Reserved	`float`	61
Field 12	Reserved	`float`	65
Field 13	Course	`float`	69
Field 14	HRMS	`float`	73
Field 15	VRMS	`float`	77
Field 16	HDOP	`float`	81
Field 17	VDOP	`float`	85
Field 18	Number of satellites	`uint8_t`	89
Field 19	Fix status	`uint8_t`	90
Field 20	Differential age	`uint8_t`	91
Checksum	Sum of bytes 1–91	`uint16_t`	92
Tail	`e`	`char`	94
Tail	`d`	`char`	95

Table 5. GNSS raw data message, data type g, 95 bytes.

3.3 Binary Euler Angle Data

The Euler angle message contains the Euler angle output (in degrees) and the Euler angle variances, that is, the rotation from the body frame to the reference frame, without any compensation of the installation misalignment angle.

Field	Data	Type	Offset
Header	`f`	`char`	1
Header	`m`	`char`	2
Header	`i`	`char`	3
Data type	`e`	`char`	4
Field 1	`HHMMSS.SS`	`double`	5
Field 2	Roll	`float`	13
Field 3	Pitch	`float`	17
Field 4	Heading	`float`	21
Field 5	Roll variance	`float`	25
Field 6	Pitch variance	`float`	29
Field 7	Heading variance	`float`	33
Field 8	Roll–pitch covariance	`float`	37
Field 9	Roll–heading covariance	`float`	41
Field 10	Pitch–heading covariance	`float`	45
Checksum	Sum of bytes 1–48	`uint16_t`	49
Tail	`e`	`char`	51
Tail	`d`	`char`	52

Table 6. Euler angle message, data type e, 52 bytes.

3.4 Binary NAVI Data

The internal Kalman filter fuses the MEMS IMU raw data with the GNSS data and outputs the result in the NAVI message protocol. The message contains the estimated latitude, longitude and height, the NED velocities (north-east-down, m/s), the Euler angles (radians), the gyro biases (rad/s) and the accelerometer biases (m/s²). Using the Euler angles and a specified mapping vector, the final coordinates can be mapped onto any point of the RTK pole, which is what makes tilt survey possible. The protocol uses little-endian byte order.

NOTE The Euler angles in the NAVI message include the compensation of the installation misalignment angle.

Field	Data	Type	Offset
Header	`f`	`char`	1
Header	`m`	`char`	2
Header	`i`	`char`	3
Data type	`n`	`char`	4
Field 1	`HHMMSS.SS`	`double`	5
Field 2	Latitude	`double`	13
Field 3	Longitude	`double`	21
Field 4	Height	`double`	29
Field 5	North velocity	`float`	37
Field 6	East velocity	`float`	41
Field 7	Vertical velocity (down)	`float`	45
Field 8	Roll	`float`	49
Field 9	Pitch	`float`	53
Field 10	Yaw	`float`	57
Field 11	Position quality	`float`	61
Field 12	Accelerometer bias X axis	`float`	65
Field 13	Accelerometer bias Y axis	`float`	69
Field 14	Accelerometer bias Z axis	`float`	73
Field 15	Gyro bias X axis	`float`	77
Field 16	Gyro bias Y axis	`float`	81
Field 17	Gyro bias Z axis	`float`	85
Field 18	Sensor temperature	`float`	89
Field 19	STATUS	`uint32_t`	93
Checksum	Sum of bytes 1–96	`uint16_t`	97
Tail	`e`	`char`	99
Tail	`d`	`char`	100

Table 7. NAVI message, data type `n`, 100 bytes.

3.5 ASCII NAVI Data

In addition to the binary format described above, the NAVI message can also be output in the following ASCII format:

```
$GPFMI,UTC,LAT,LON,HGT,ROLL,PITCH,YAW,HRMS,NSAT,AGE,NAVMOD,STATUS*CHECKSUM
```

The ASCII message has the same meaning as the binary NAVI message. The checksum is calculated with the NMEA-0813 checksum algorithm. HRMS, NSAT, AGE and NAVMOD are the GNSS position details obtained by the IMU.

3.6 The STATUS Word in the NAVI Message

The STATUS word is defined as follows:

No.	Name	Bit	Description
1	Finit	0X00000001	Filter not initialised
2	Ready	0X00000002	Filter convergence complete
3	Inaccurate	0X00000004	Filter convergence in progress
4	TiltReject	0X00000008	Tilt angle exceeds the threshold
5	GnssReject	0X00000010	Poor RTK data quality
6	FReset	0X00000020	Filter reset
7	FixRlsStage1	0X00000040	Installation angle estimation, stage 1
8	FixRlsStage2	0X00000080	Installation angle estimation, stage 2
9	FixRlsStage3	0X00000100	Installation angle estimation, stage 3
10	FixRlsStage4	0X00000200	Installation angle estimation, stage 4
11	FixRlsOK	0X00000400	Installation angle estimation complete
12	Direction1	0X00002000	Swing the RTK pole to initialise direction 1
13	Direction2	0X00004000	Swing the RTK pole to initialise direction 2
14	GnssLost	0X00010000	RTK data invalid or missing
15	FInitOk	0X00020000	Filter initialisation complete
16	PPSReady	0X00040000	PPS signal detected
17	SyncReady	0X00080000	Module time synchronised with PPS
18	GnssConnect	0X00100000	Serial connection to the RTK established

Table 8. STATUS word bit definitions (NAVI field 19 and the ASCII STATUS field).

NOTE 7–11: The module supports calibration compensation for the RTK pole; fields 7 to 11 indicate the calibration status.

12–13: The pole swing direction fields are used during initialisation and prompt the operator to rotate the RTK pole by 90°. If the device is not factory calibrated, then once the IMU is time-synchronised and the RTK has a fixed solution, swing the RTK pole back and forth for 1–3 seconds, rotate it by 90° and swing again for 1–3 seconds to complete the initialisation.

16–18: The IMU module must be time-synchronised with the RTK receiver. The host MCU achieves this by configuring the RTK receiver to output a 1PPS signal (falling edge by default) together with the associated RTK data.

4 Tilt Survey Mode — Configuration Example

For tilt survey mode, send the following commands after start-up.

```
AT+NAVI_OUTPUT=UART1,ON
AT+LEVER_ARM2=0,0,0
AT+CLUB_VECTOR=0,0,1.855
AT+GNSS_CARD=HEMI
AT+WORK_MODE=152
AT+SAVE_ALL
```

NOTE A. If the FMI IMU module is mounted on the back of the GNSS receiver — that is, facing down instead of up — send `AT+INSTALL_ANGLE=180,0,0` before the save command.

B. The communication interface, the GNSS receiver type and LEVER_ARM2 and CLUB_VECTOR (RTK pole) must be set according to the specific hardware implementation.

C. After the RTK tilt survey system is switched on, the GNSS receiver outputs an integer fixed solution and the PPS signal. Hold the tip of the RTK pole still on the ground. Swing the RTK pole back and forth once or twice; the IMU module then switches the position status to “initialisation complete” and the tilt survey can begin.

D. If the installation angle is not factory calibrated, the RTK pole must be rotated by 90° at the first initialisation, as prompted by the IMU. In continuous operation the 90° rotation is not required.

5 Agricultural Heading Mode — Description and Example

When an agricultural machine rolls or pitches by X° on a slope, its antenna does not merely shift relative to the ground but also describes an arc around the rear axle. With the known lever arm vector passed to our IMU, the system can use the attitude matrix to calculate exactly how the antenna movement changes the position of the rear axle. Otherwise the error can grow several times larger, wasting crop protection agents or crushing plants.

In agricultural heading mode, align the X axis of the IMU with the direction of travel of the machine. If the installation angle does not point forward, adjust it with the `AT+INSTALL_ANGLE` command. The lever arm and wheel arm parameters must also be set according to the actual design. The basic configuration for agricultural heading mode is described below.

> **NOTE** In the NAVI message output in agricultural heading mode, the accelerometer and gyro bias fields are replaced by the algorithmically compensated acceleration and angular rate, which are intended primarily for agricultural machine control.

Example of the basic configuration steps in agricultural heading mode:

```
AT+NAVI_OUTPUT=UART1,ON
AT+LEVER_ARM2=X,Y,Z
AT+WHEEL_ARM=X,Y,Z
AT+GNSS_CARD=HEMI
AT+WORK_MODE=72
AT+SAVE_ALL
```

6 Dead Reckoning Mode — Description and Example

Dead reckoning mode turns a satellite-dependent GNSS system into an inertial navigation system (INS) with graceful degradation: if the satellite signal is lost, continuous high-rate navigation is maintained in underground car parks, tunnels and urban canyons.

In dead reckoning mode, align the X axis of the IMU with the direction of travel of the vehicle. If the installation angle does not point forward, adjust it with the `AT+INSTALL_ANGLE` command. The lever arm and wheel arm parameters must also be set according to the actual design. The basic configuration is described below.

> **NOTE** In the NAVI message output in dead reckoning mode, the accelerometer and gyro bias fields are replaced by the algorithmically compensated acceleration and angular rate, which are intended primarily for agricultural machine control.

Example of the basic configuration steps in dead reckoning mode:

```
AT+NAVI_OUTPUT=UART1,ON
AT+INSTALL_ANGLE=X,Y,Z
AT+LEVER_ARM2=X,Y,Z
AT+WHEEL_ARM=X,Y,Z
AT+GNSS_CARD=HEMI
AT+WORK_MODE=13
AT+SAVE_ALL
```